



Data Analytics Assessment

Vahan Lead Conversion Analysis

Cohort Performance, SQL Aggregation and FT prediction

Description

An Analysis of lead -source performance and FT conversion designed to identify high performing cohort, aggregate lead-level data into actionable business metrics, and investigate factors associated with FT conversion.

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Executive Summary

Vahan Sources candidate leads and uses calling operations to identify leads that ultimately convert to FT. This analysis evaluates the relationship between the lead source and FT conversion and develops a ML model to investigate factor associated with FT

1. Cohort Performance

Lead sources were evaluated using FT conversion rate as the primary metrics, with lead volume and FT conversion count as the supporting measures. The top three cohort was identifies based on this frame work

2. Aggregate Analysis

The lead-level dataset was aggregated at the lead-source level, providing total leads, FT conversion rate for each cohort

3. ML Analysis

Logistic Regression was selected as the primary classification model because FT is a binary outcome and the assessment requires interpretable factor influencing FT .This model was evaluated using precision, recall,F1-Score,ROC-AUC and a confusion matrix.

Business Problem and Data

Business Problem:

Vahan sources leads from multiple channel and contacts these candidates through a calling process. The Objective is to understand which lead sources generate stronger conversion outcome and which candidate/lead characteristics are associated with the likelihood of FT

Assessment Question:

1. Identify the 3 best lead-source cohort and define the appropriate performance metric
2. Aggregate the raw data at an appropriate business level and provide the SQL query
3. Build an ML model to identify factors associated with FT and provide a confusion matrix

Data Overview

Metric	Value
Total Lead	18,198
FT Leads	17
Non-FT Leads	18,181
FT Rate	0.093%
Cohort Dimension	Lead Source

Data Preparation:

The dataset was checked for duplicate record, missing value, data types and target consistency. Data field were converted to appropriate datetime formats and relevant data-quality checks were completed before analysis

Cohort Analysis

Cohort Definition

The cohort definition was defined as lead _source, as the business objective is to determine which sourcing channel generate leads with stronger FT conversion performance

Metric Selection:

Primary Metrics: FT conversion rate

$$FT\ Conversion\ Rate = \frac{FT\ Conversion}{Total\ Leads} * 100$$

Conversion rate allows cohort with different lead volume to be compared on a common efficiency measure. FT count and total lead volume were considered as supporting metrics to avoid overvaluing small cohort

TOP 3 Cohort:

...	lead_source	total_leads	ft_after_upload	upload_ft_rate	
15	Single Referral > 7 days- 24th Jul	1500	14	0.933333	
11	Khanna- 2W 26th Jul	1546	14	0.905563	
13	PreOb-Ob Fees Paid 29th Jul (set 1)	1483	7	0.472016	

The top-performing cohorts indicated the lead sources that have historically generated stronger FT conversion efficiency. These sources can be considered for closer monitoring and potential prioritization, subject to sufficient lead volume

SQL Analysis

Aggregation Level

The raw candidate-level data was aggregated at the lead source level matching the cohort definition. This level provides a direct comparison of sourcing channel performance

```
query = """
SELECT
    lead_source,
    COUNT(*) AS total_leads,
    SUM(ft_after_first_attempt) AS ft_candidates,
    ROUND(
        100.0 * SUM(ft_after_first_attempt) / COUNT(*),
        3
    ) AS ft_conversion_rate
FROM raw_leads
GROUP BY lead_source
ORDER BY ft_conversion_rate DESC;
"""
```

```
sql_result = pd.read_sql_query(query, conn)
```

```
sql_result
```

	lead_source	total_leads	ft_candidates	ft_conversion_rate
0	Khanna- 2W 26th Jul	1546	6	0.388
1	PreOb-Ob Fees Paid 29th Jul (set 2)	1558	5	0.321
2	PreOb-Ob Fees Paid 29th Jul (set 1)	1483	2	0.135
3	Single Referral > 7 days- 24th Jul	1500	2	0.133
4	OLX - Ashwin - 2W - 17 Jul	5182	2	0.039
5	Khanna AI	886	0	0.000
6	JobHai - Khanna - Riders - 17 Jul	3	0	0.000
7	JobHai - Khanna - Riders - 16 Jul	2	0	0.000
8	AI Connected but not Connected by TC- Set 2	1193	0	0.000
9	AI Connected but not Connected by TC- Set 1	1480	0	0.000
10	AI Connected band Not Interested	2137	0	0.000
11	50K 2W5 - Khanna - 2W - 17 Jul	1	0	0.000
12	50K 2W4 - Khanna - 2W - 17 Jul	1	0	0.000
13	2W3W - 3WEV - Khanna - 3W - 17 Jul	201	0	0.000
14	2W3W - 3WCNG - Khanna - 3W - 17 Jul	530	0	0.000

Machine Learning

FT prediction- Objective

The objective of the ML analysis is to estimate the likelihood of the FT identity variables associated with FT conversion

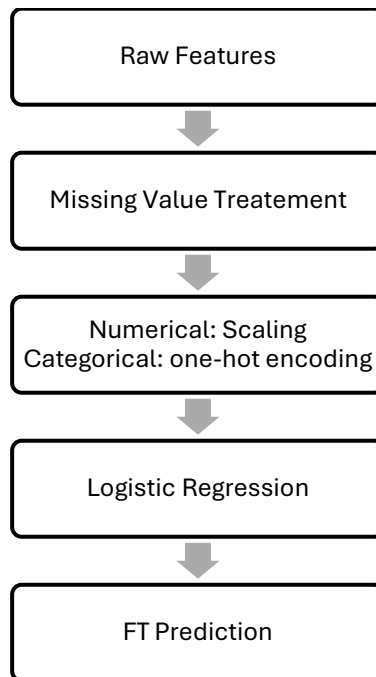
Target: FT_after_first_attempt

Model

Logistic Regression

Logistic Regression was selected as the primary model because FT is a binary outcome and the assessment explicitly requires understanding factors that influences FT. Its coefficient provide an interpretable indication of the direction of association between predictor ad FT

Preprocessing



Preprocessing was implemented within a pipeline to ensure transformation were learned from training and applied consistently to the test data

Model Performance

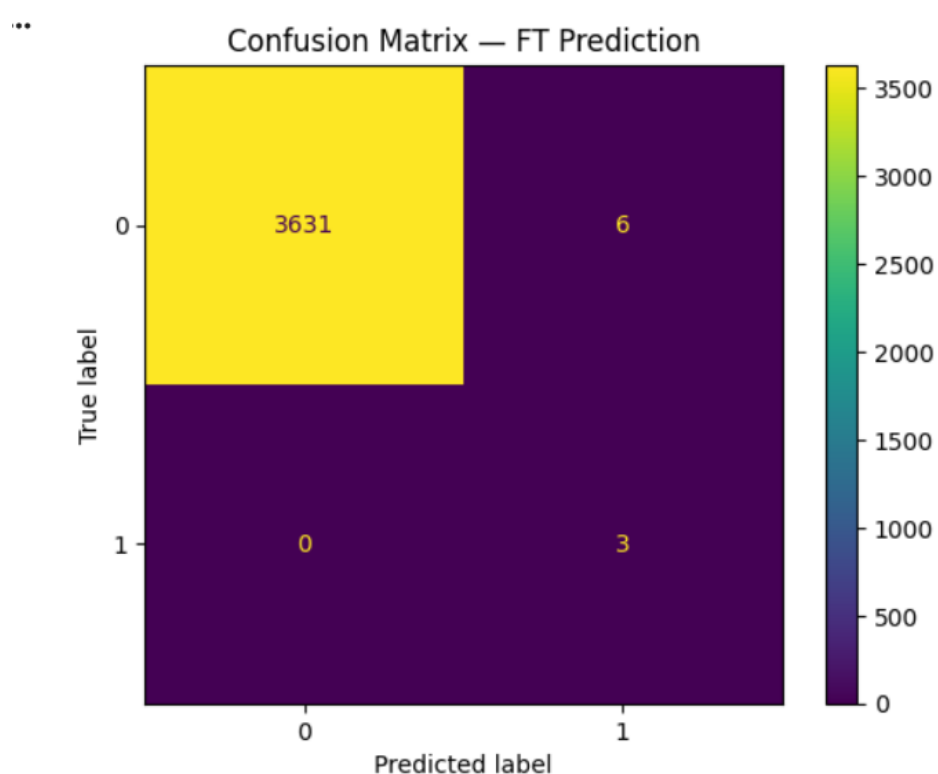
...

	Metric	Score	
0	Accuracy	0.998352	
1	Precision	0.333333	
2	Recall	1.000000	
3	F1 Score	0.500000	
4	ROC-AUC	0.999450	

Accuracy is high largely because FT is extremely rare. The model correctly identified all three FT cases in the Test set ,resulting in 100% recall, but generated 6 false positives resulting in 33.3% precision

Confusion Matrix and Model Interpretation

Confusion Matrix:



The Model identified all three observed cases in the test set and produced no false negatives. Though, six non FT leads were incorrectly classified as FT. This represents a recall oriented model as it avoids observed FT cases at the cost of additional false positive leads requiring attention

Factors Associated with FT

During the feature interpretation variables representing candidates identifier and event occurring during calling process were identified as potential sources of information leakage. These variables should not be interpreted as genuine pre-call predictor of FT

The final interpretation should therefore focus only on features that are available at the time the lead is received.

Model Limitation

Analytical Limitation:

1. Severe Class Imbalance

Only 17 of 18198 observation are FT- Positive, representing approximately 0.093% of the dataset. Therefore the accuracy is not appropriate standalone measure of model quality

2. Limited Positive sample

The small number of FT-positive observations means that performance and feature association should be interpreted cautiously

Final Finding and Recommendation

Finding 1: Lead source

The strongest lead source cohorts were identified using FT conversion rate as the primary metrics, supported by lead volume and FT count

Finding 2: Aggregation

Lead-source level aggregation provides an appropriate business view for comparing sourcing channel efficiency

Finding 3: ML

Logistic Regression provides an interpretable framework for estimating FT likelihood and identifying associated factors

Finding 4: Model Performance

The model achieve 100% recall on test set but only 33.3% precision demonstrating the trade-off between potential FT leads and generating false positives

Recommendation

Vahan can use the cohort analysis to monitor sourcing channel efficiency and prioritize high performing lead sources. A predictive lead prioritization model may be useful operationally, but it should be retained and validated on a larger set of FT positive observations using only information before calling begins